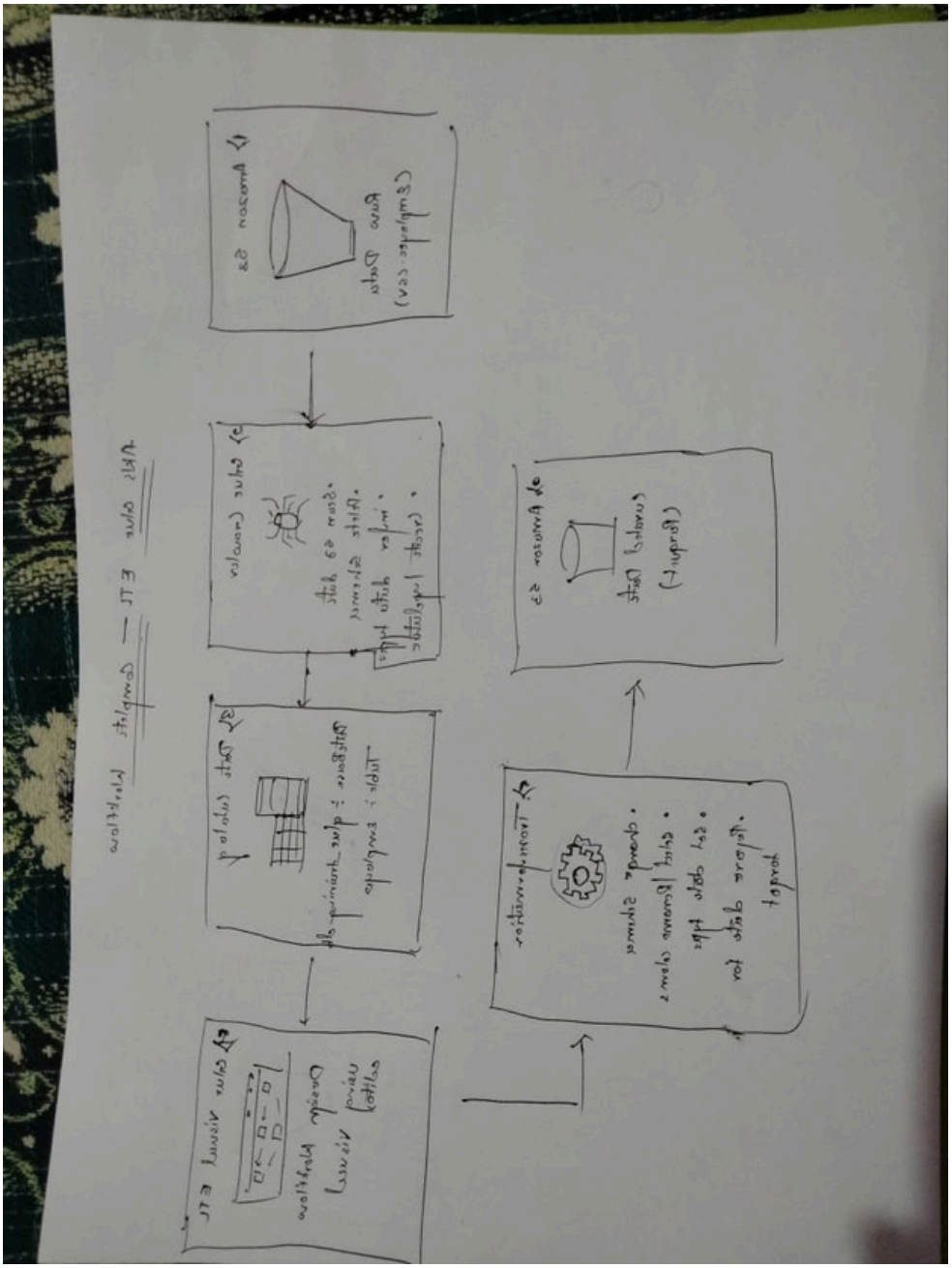
AWS CLOUD &
DATA ENGINEERING

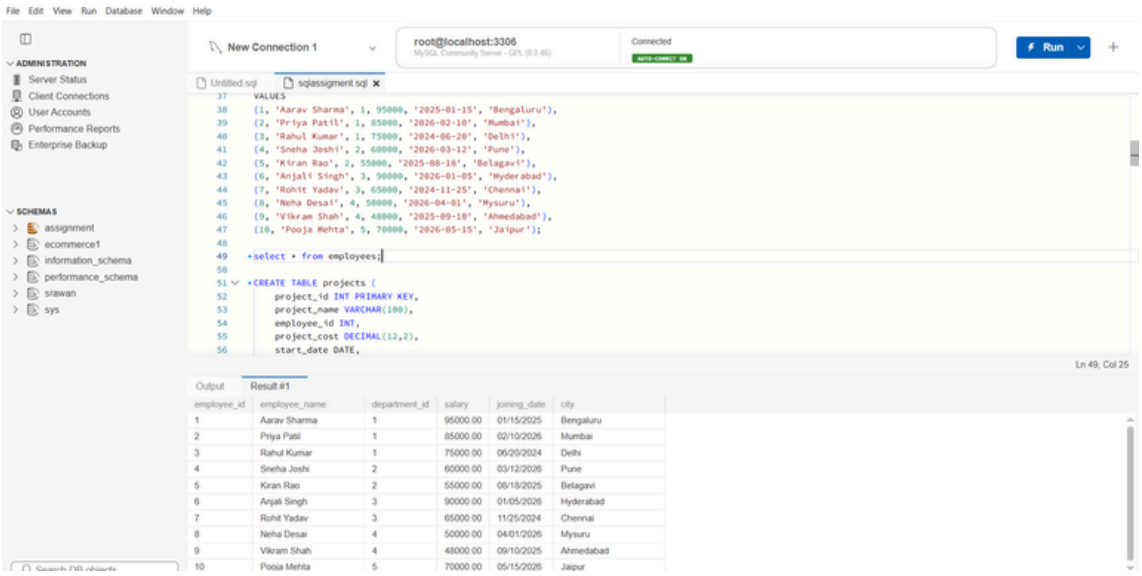
Name	SRAWAN VITHAL BUCCHE
USN / Roll Number	2JI23CS169
College Name	Jain College of Engineering, Belagavi
Batch	2023 - 2027
Branch	Computer Science Engineering
Title	AWS Cloud & Data Engineering Practical Task

Workflow

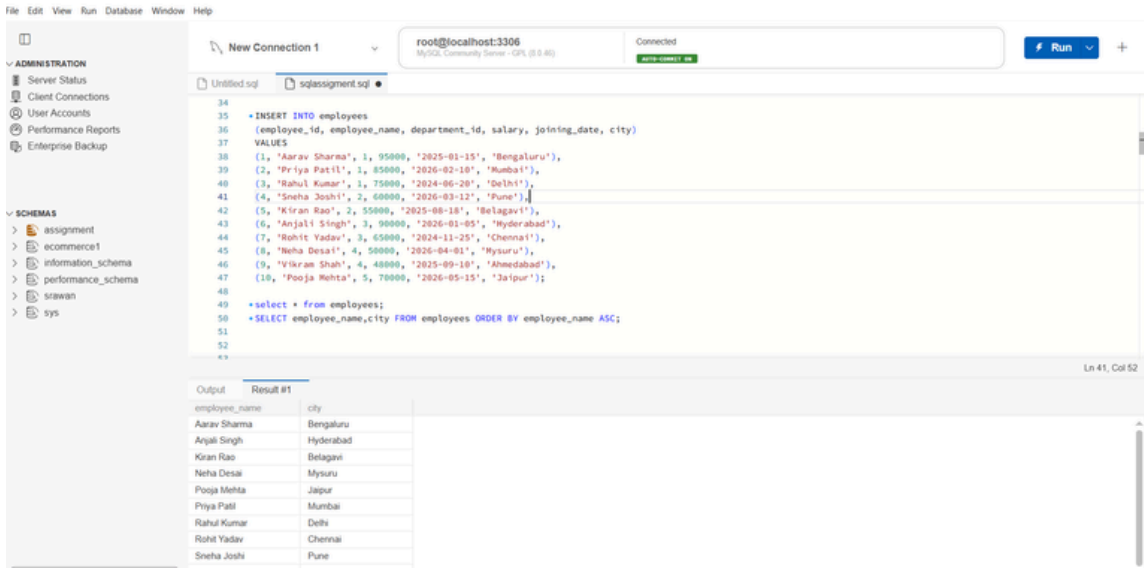


SQL PRACTISE

Display all employees ?

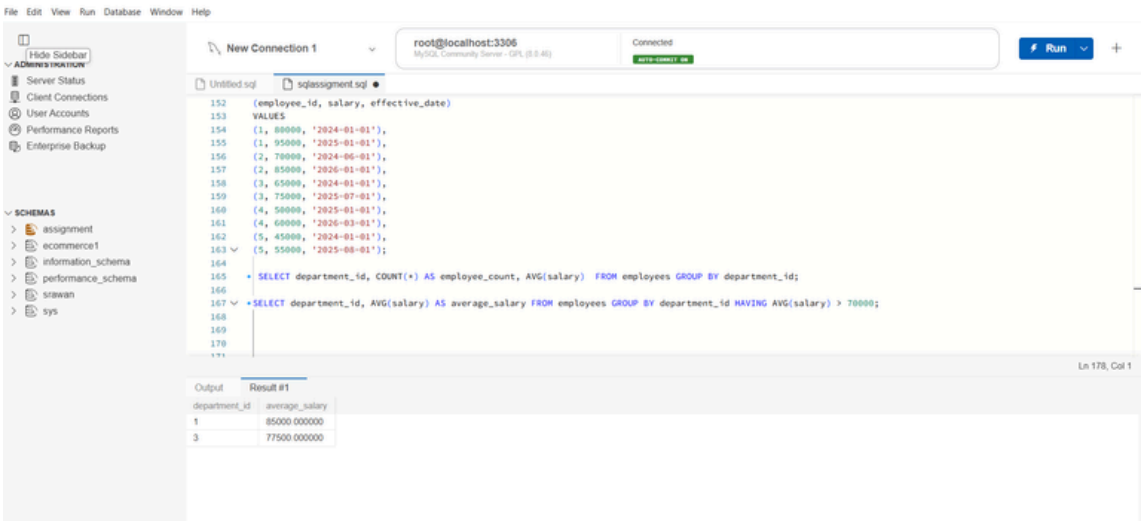


Display employees names and cities, sorted alphabetically by name.

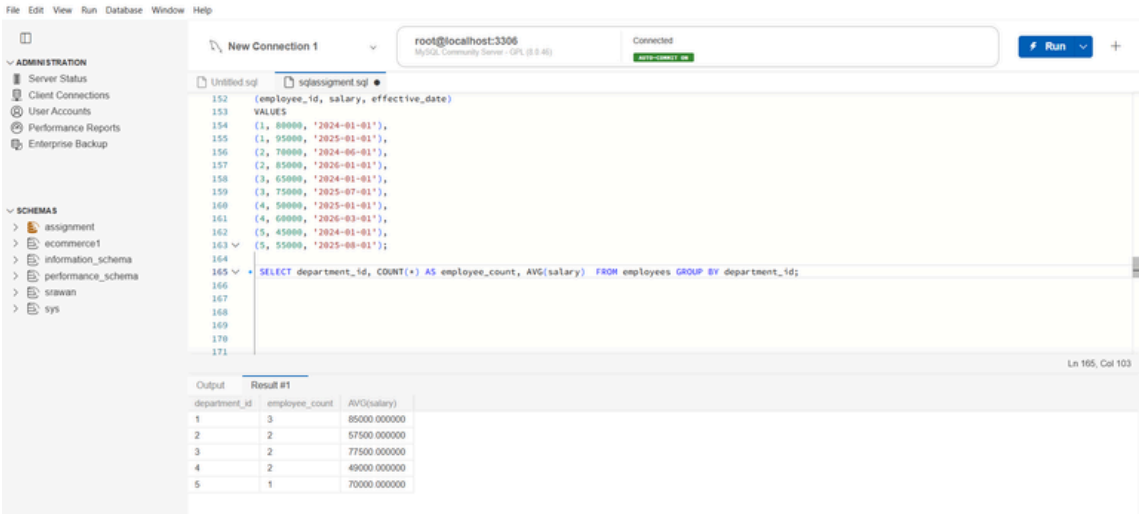


\

Find all employee name where salary is more than 70000

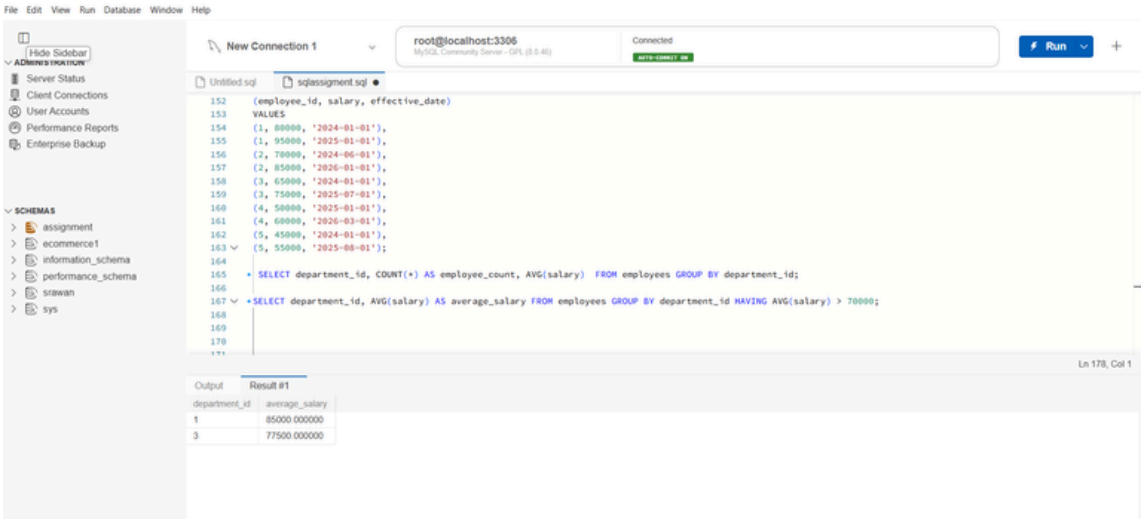


employee count and avg salary for each department

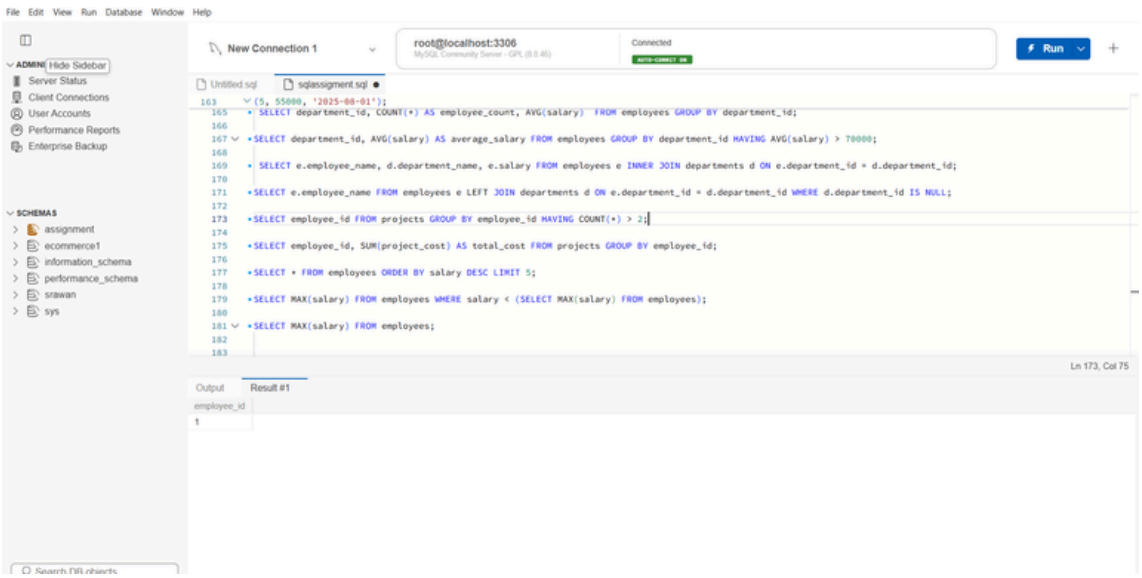


\

Return department whose Avg salary is greater than 70000

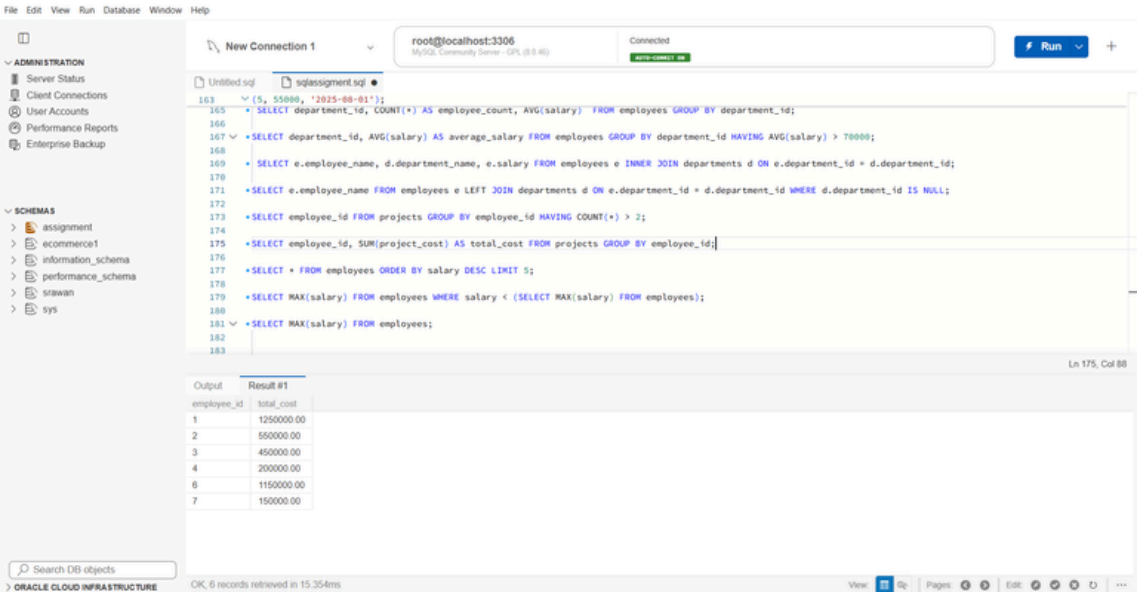


employees assigned to or more than 2 project ?

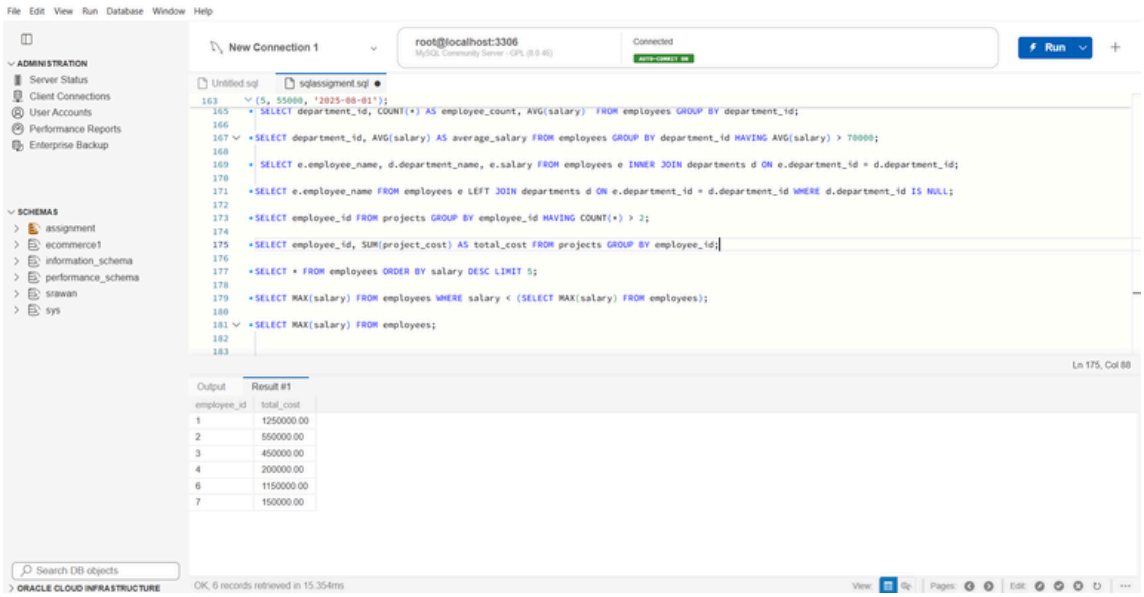


\

Calculate total project cost for each employee ?

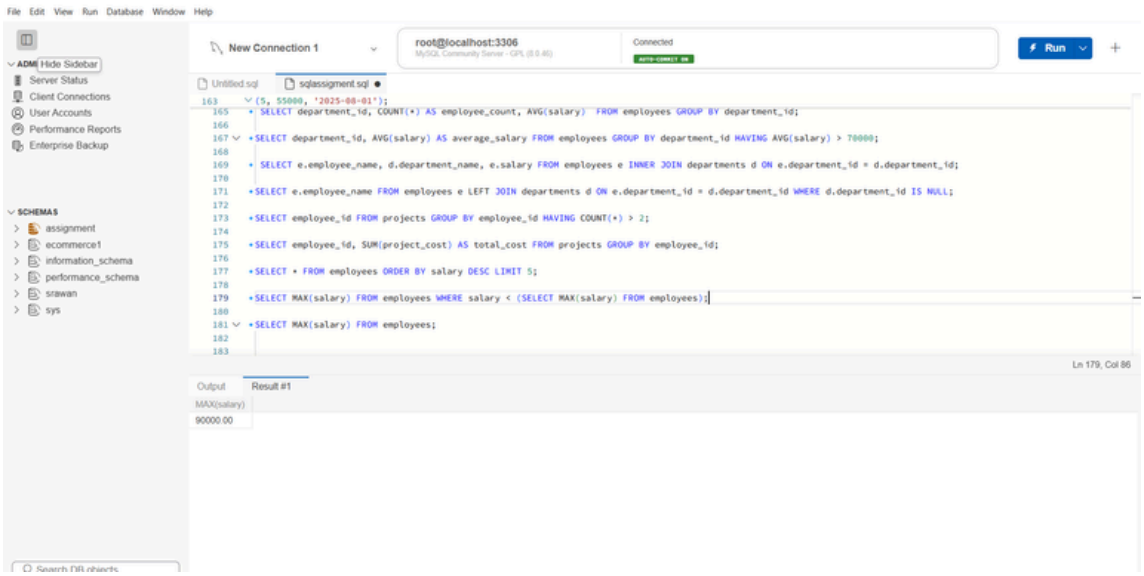


The 5 highest salaries ?

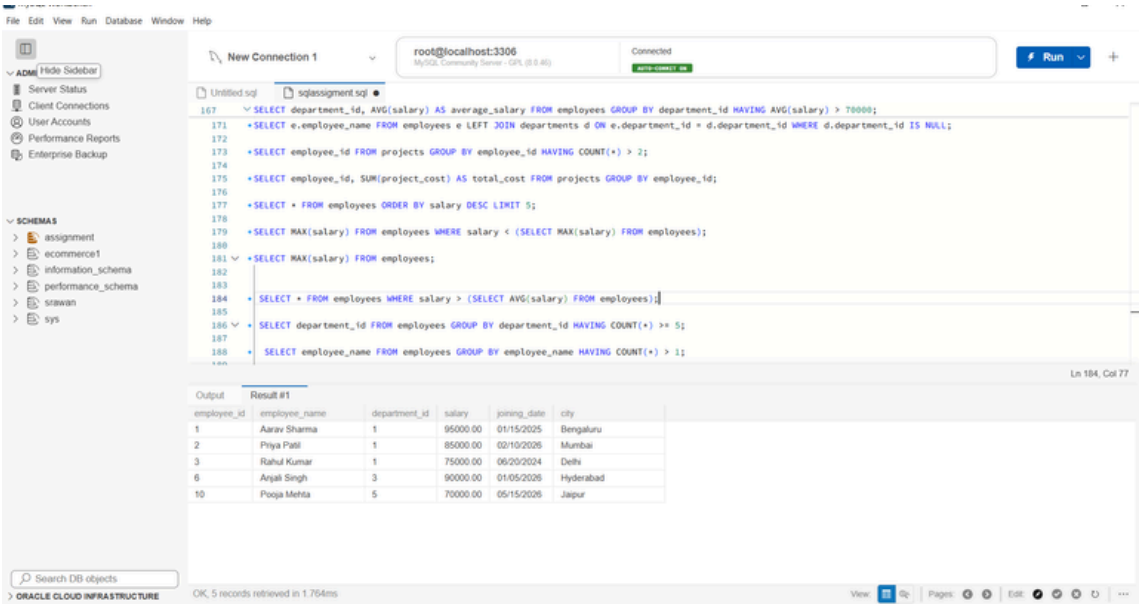


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Second-Highest Distinct Salary ?

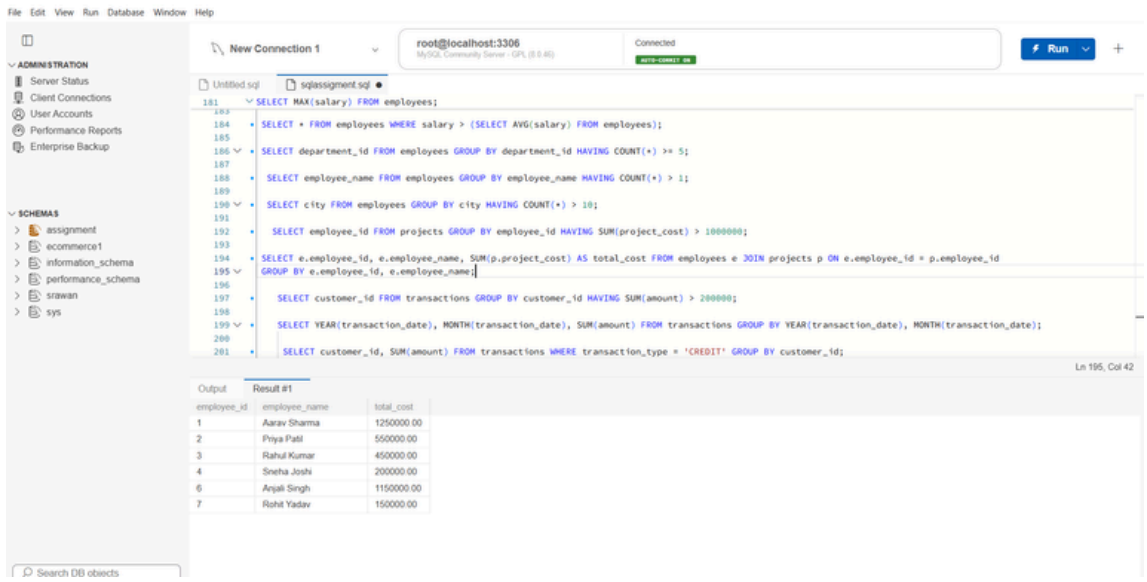


Employees above Overall Averagee ?

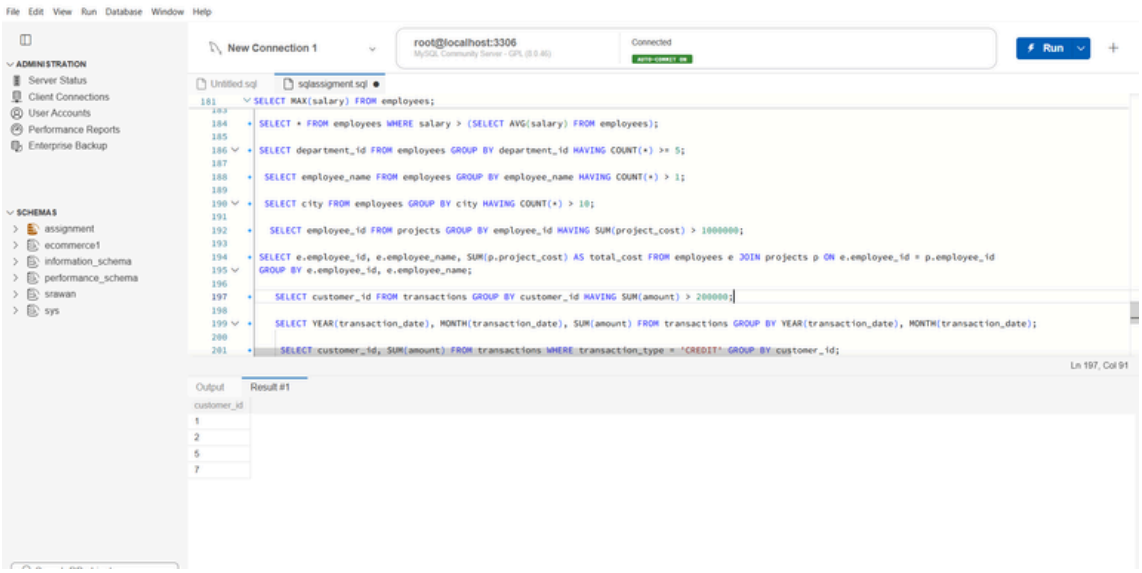


\

Each employee total project cost

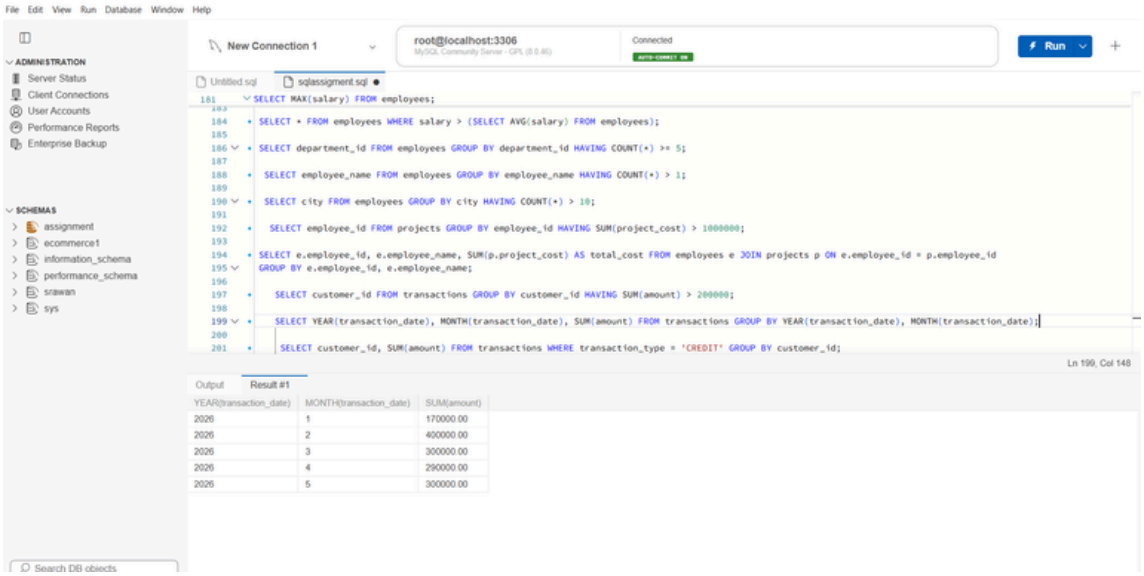


Customer whose tatal transaction amount exceeds 200000

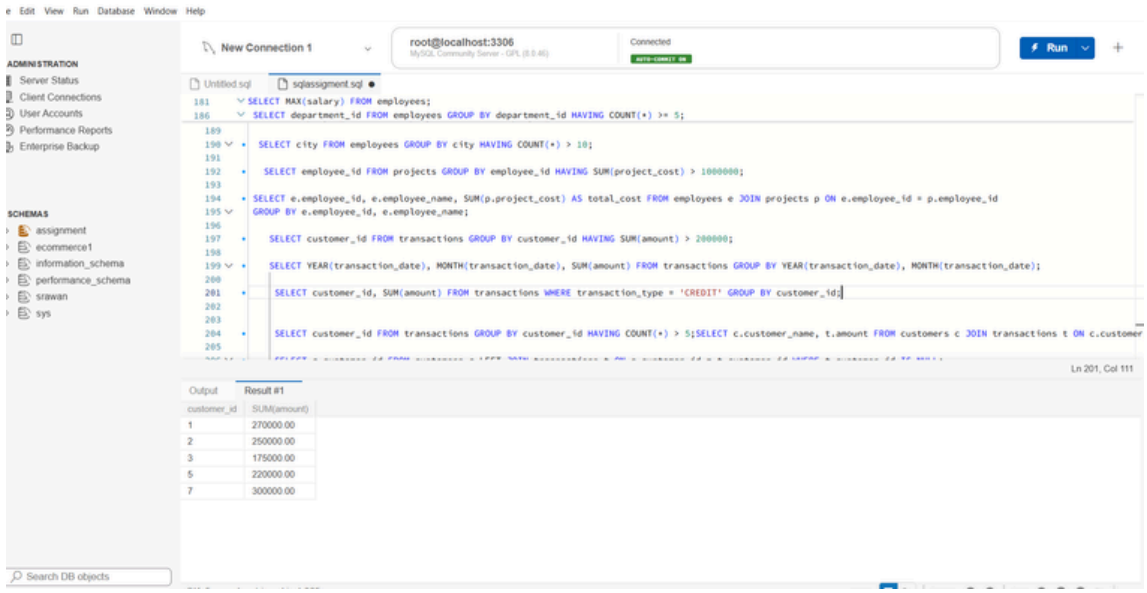


\

Calculate total transaction amount by year and month

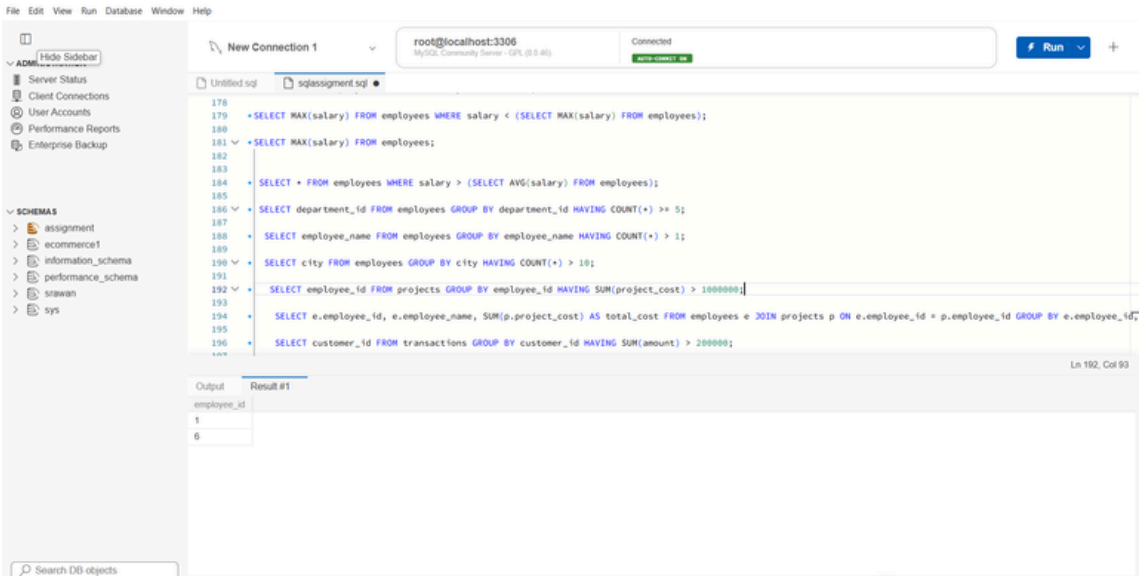


Calculate total CREDIT amount for each customer



\

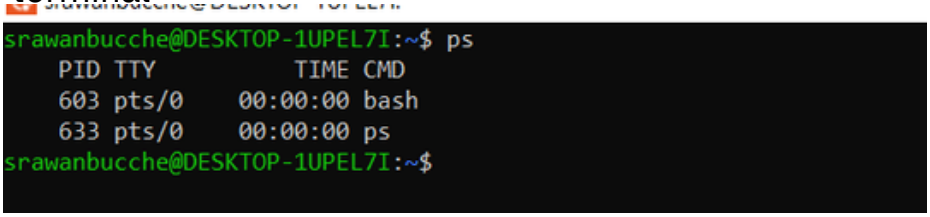
High project cost employees exceeds 1000000?



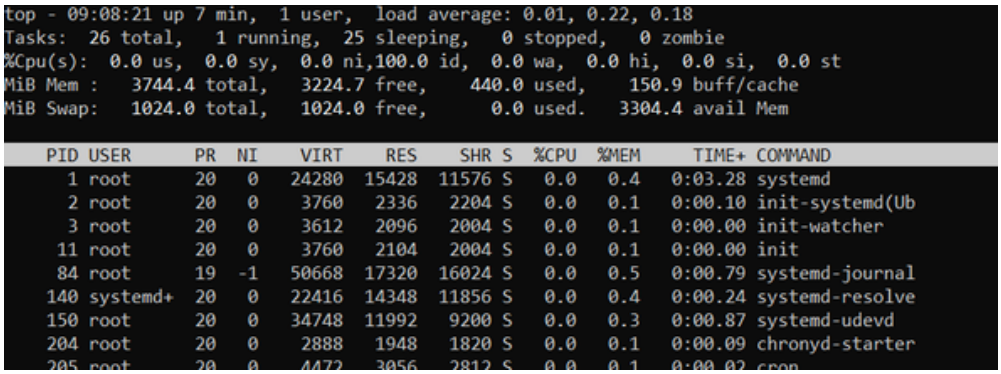
\

Linux Practice:

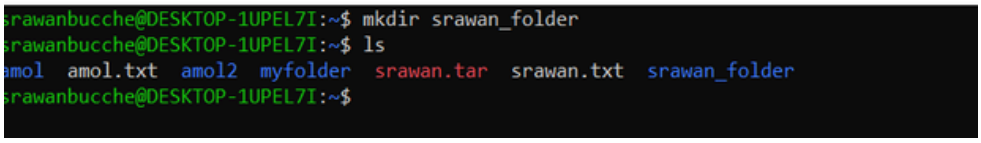
ps : Display the running processes of the current terminal



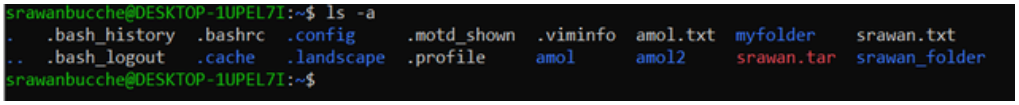
top : Display live process and memory usage



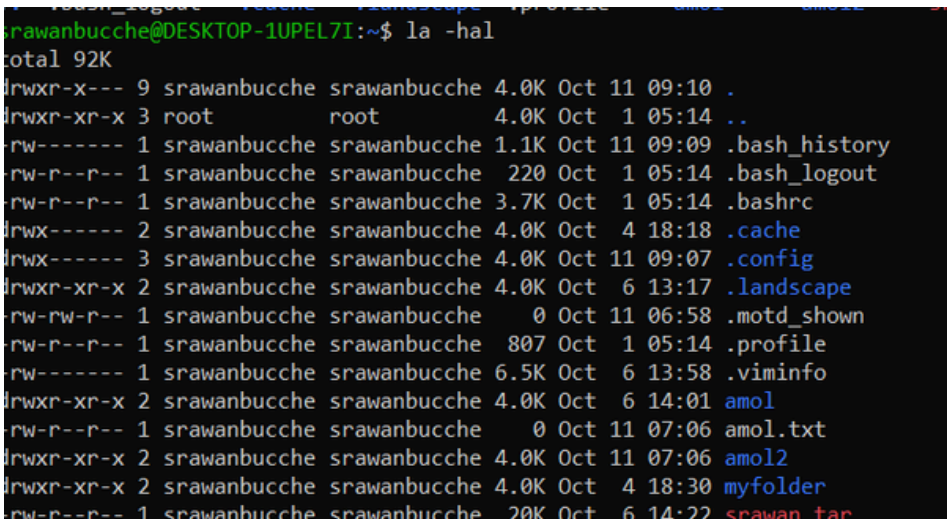
mkdir and ls : Create a directory and list it



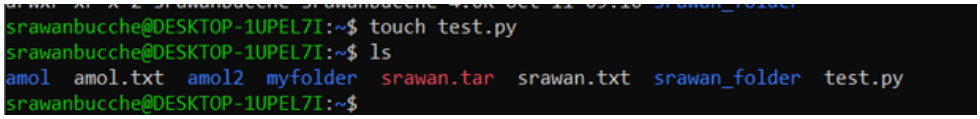
ls -a : List all files including hidden files



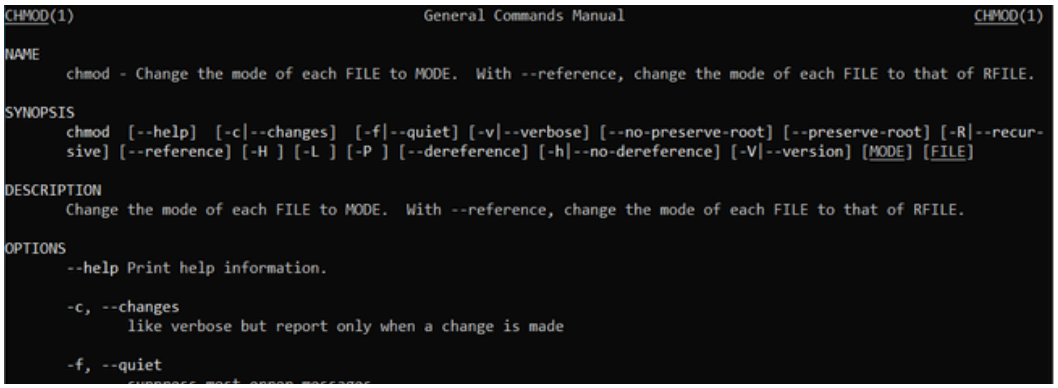
ls -hla : Long listing with permissions and human-readable sizes



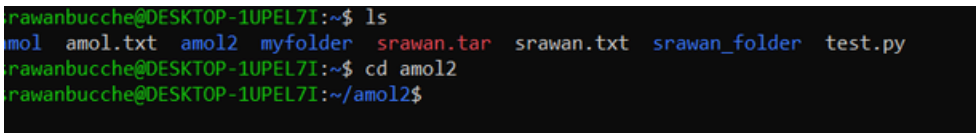
touch and ls : Create an empty file and list it



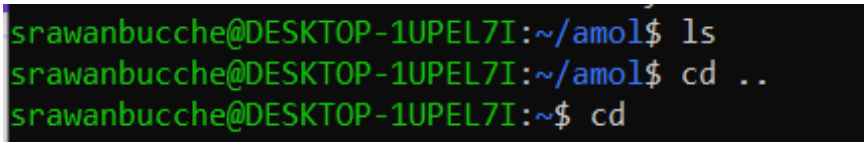
man chmod : Manual page of the chmod command



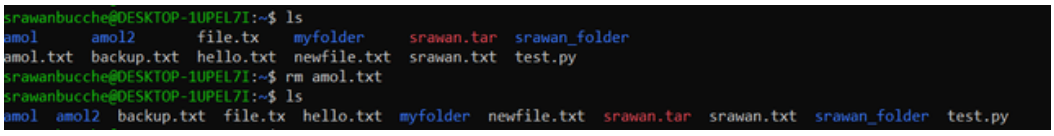
cd : Change the current directory



cd .. :move to parent Directory



rm : Remove a file



cat: Display file contents

```
srawanbucche@DESKTOP-1UPEL7I:~$ cat hello.txt
Hello linux
srawanbucche@DESKTOP-1UPEL7I:~$
```

nano : create or edit file

```
awanbucche@DESKTOP-1UPEL7I:~$ nano file.txt
awanbucche@DESKTOP-1UPEL7I:~$ cat file .txt
at: file: No such file or directory
at: .txt: No such file or directory
awanbucche@DESKTOP-1UPEL7I:~$ cat file.txt
ello linux
awanbucche@DESKTOP-1UPEL7I:~$
```

pwd : Show the present working directory

```
srawanbucche@DESKTOP-1UPEL7I:~$ pwd
/home/srawanbucche
srawanbucche@DESKTOP-1UPEL7I:~$
```

sudo cp : Copy a file to /home

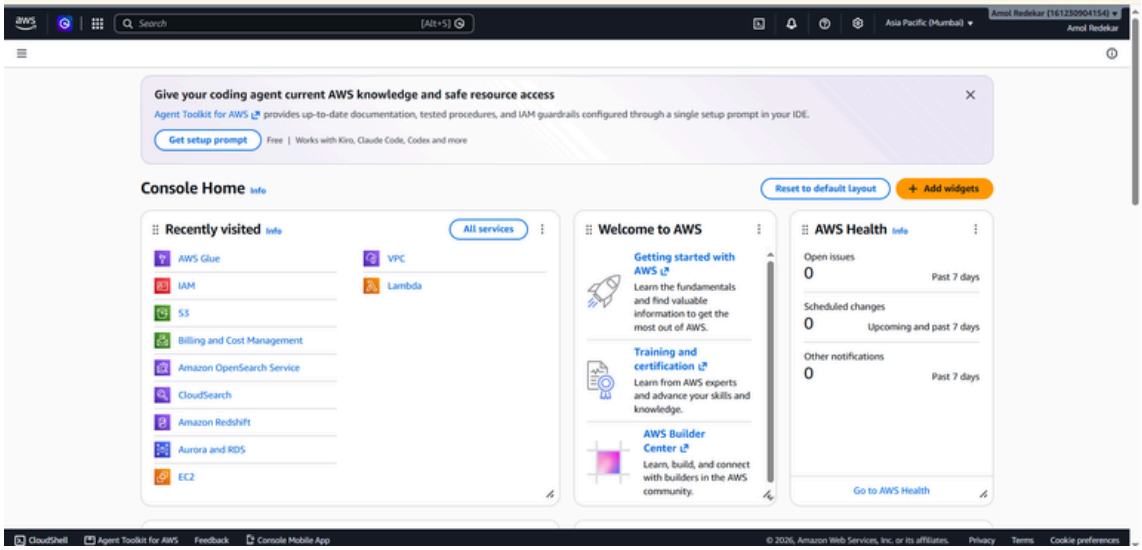
```
rawanbucche@DESKTOP-1UPEL7I:~$ cp file.txt backup.txt
rawanbucche@DESKTOP-1UPEL7I:~$ ls
mol       amol2       file.tx   hello.txt  srawan.tar  srawan_folder
mol.txt   backup.txt  file.txt  myfolder   srawan.txt  test.py
rawanbucche@DESKTOP-1UPEL7I:~$
```

sudo mv : Move a file to the root directory

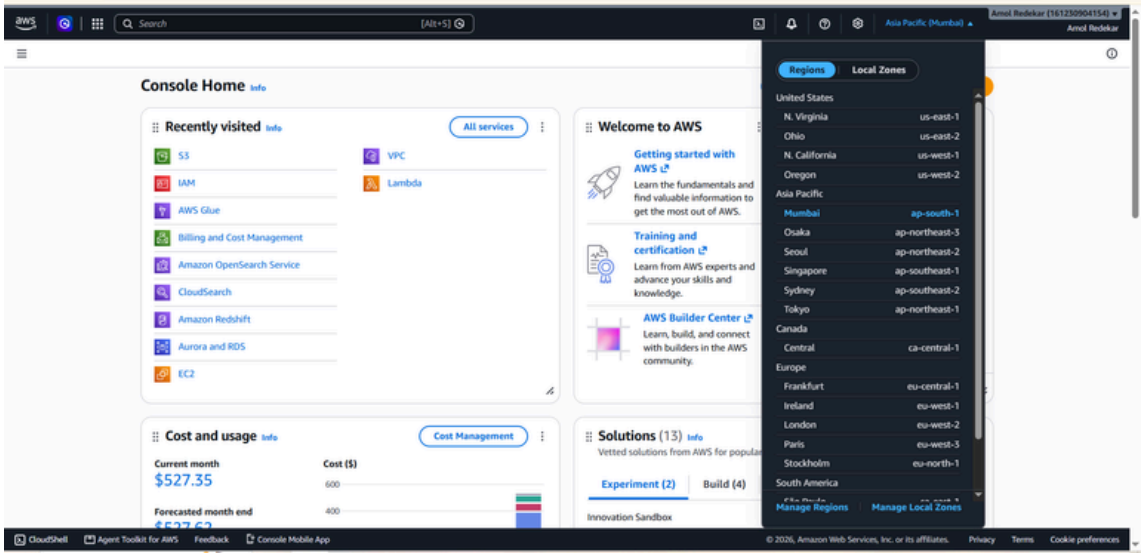
```
mv: cannot stat 'file': No such file or directory
srawanbucche@DESKTOP-1UPEL7I:~$ mv file.txt newfile.txt
srawanbucche@DESKTOP-1UPEL7I:~$ ls
amol      amol2       file.tx    myfolder     srawan.tar
amol.txt  backup.txt  hello.txt  newfile.txt  srawan.txt
srawanbucche@DESKTOP-1UPEL7I:~$
```

AWS Account

AWS Console



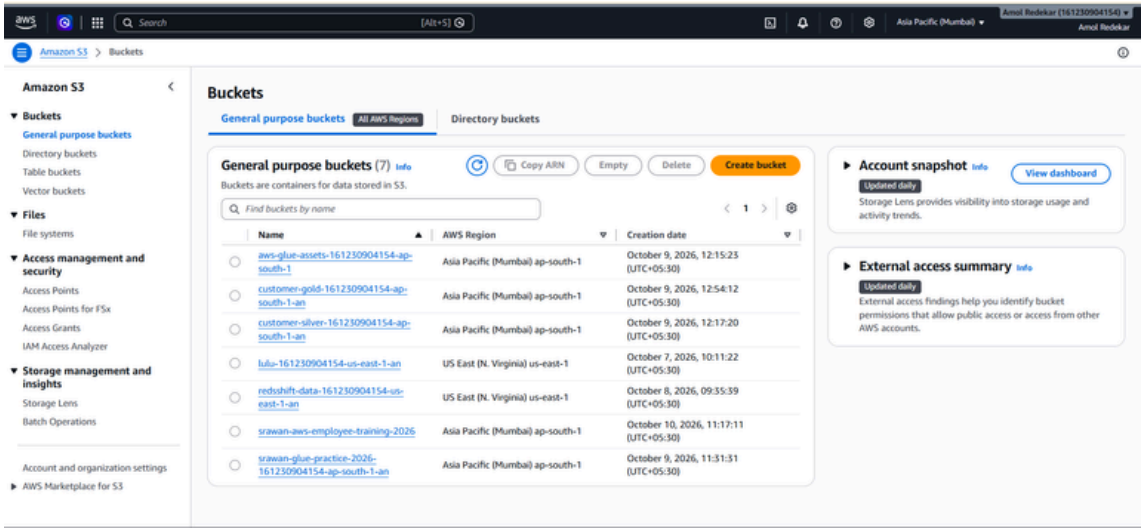
Selected AWS Region



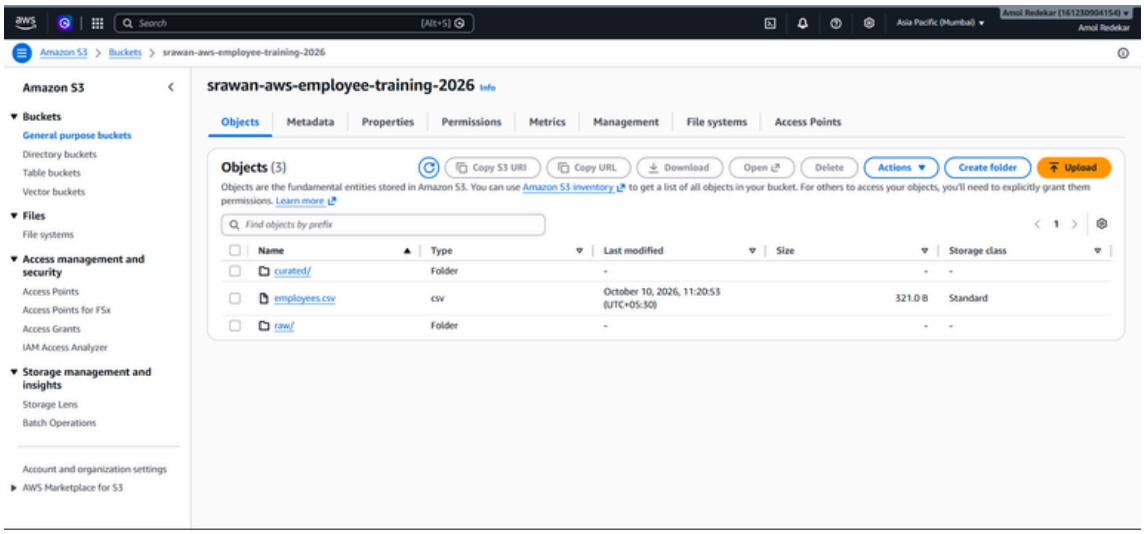
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Amazon S3

Create S3 bucket : srawan-aws-employee-training-2026



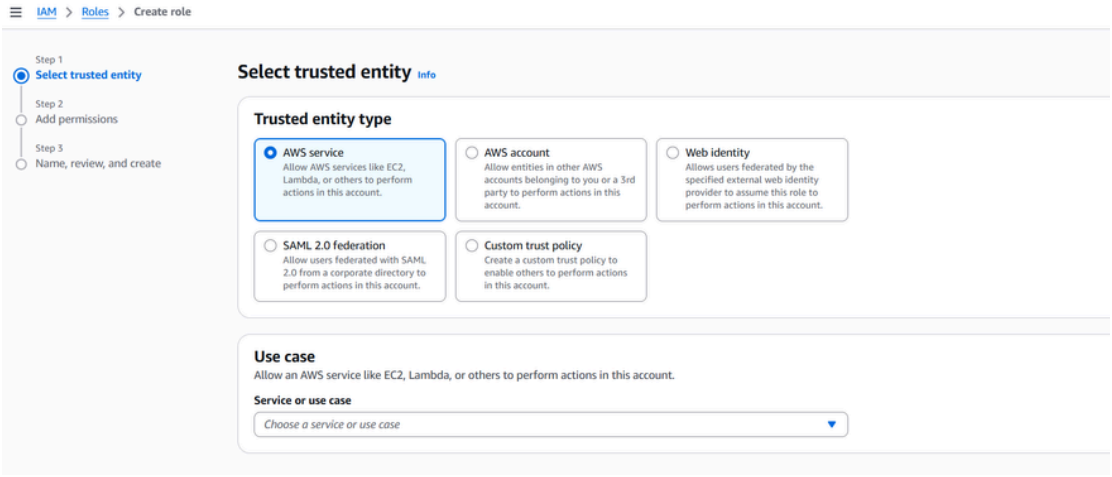
Create object : raw and curated



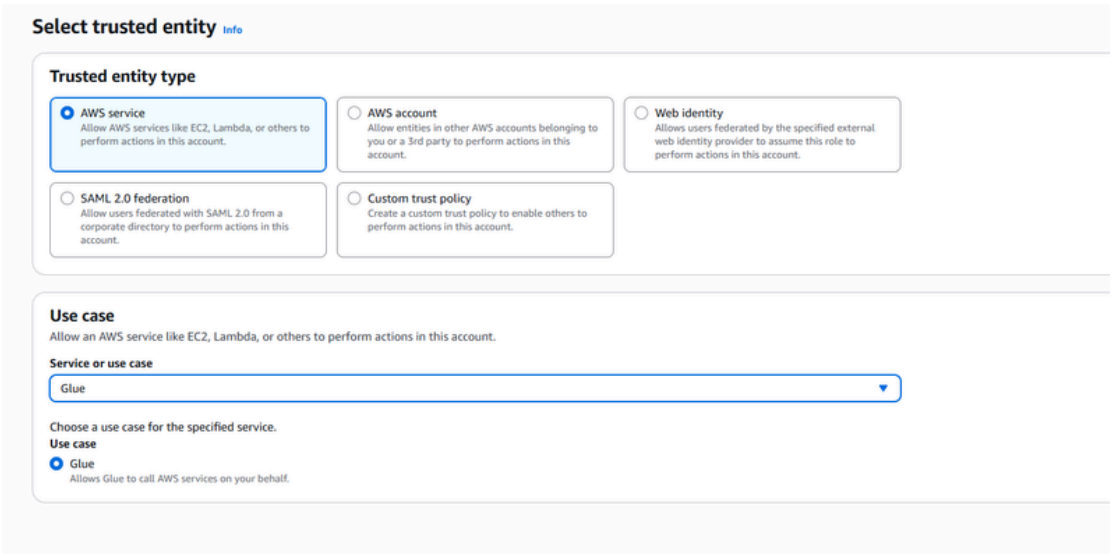
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IAM:

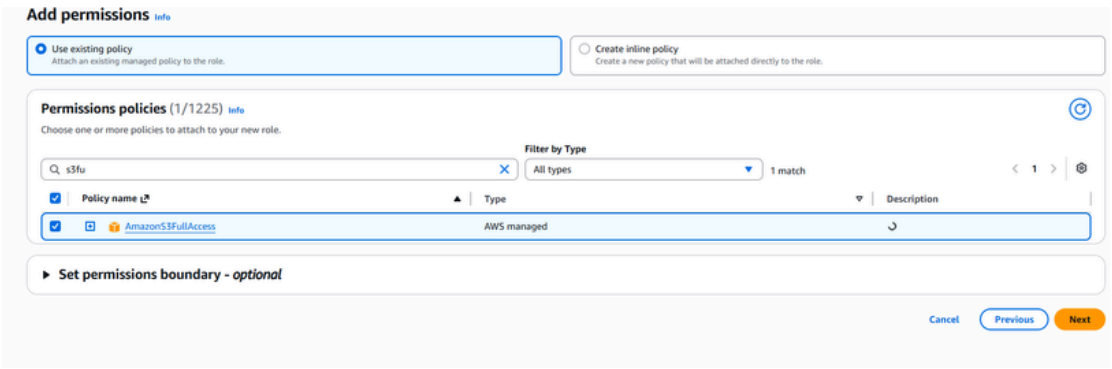
Select trusted entity as AWS service



Use case selected as Glue

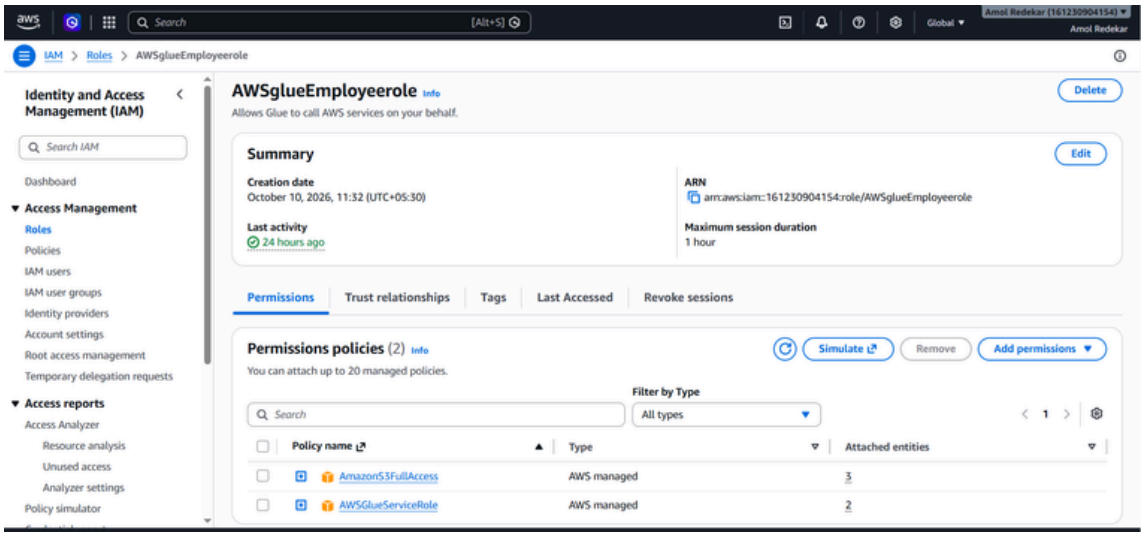


Add permissions: AmazonS3FullAccess policy selected

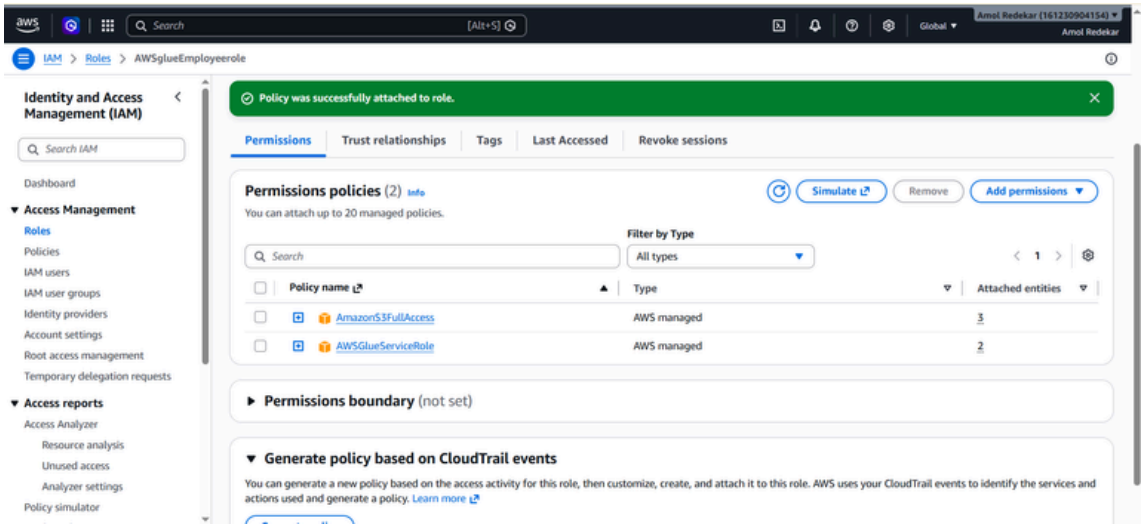


IAM

IAMRole used for AWS Glue



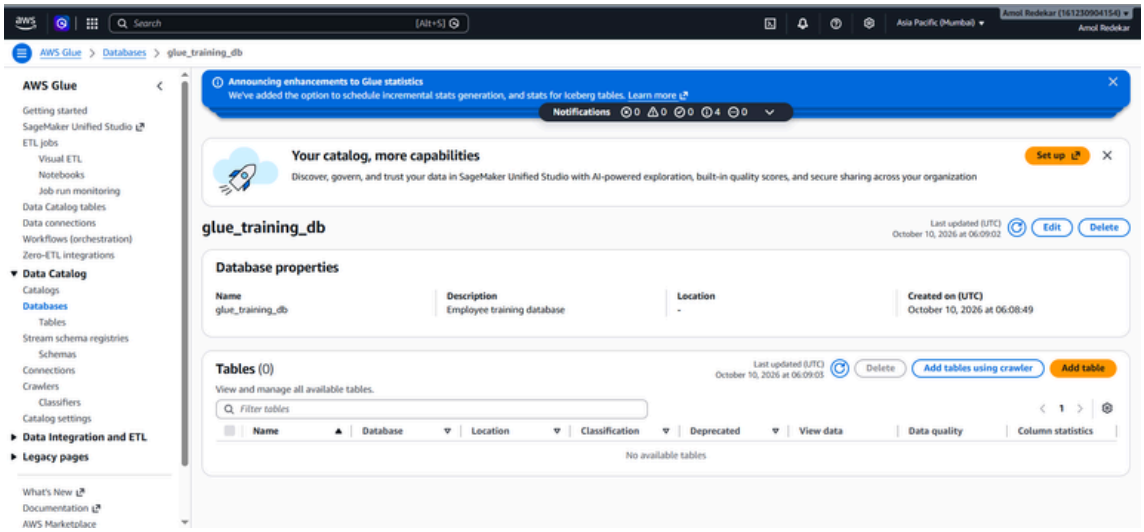
Add Policy



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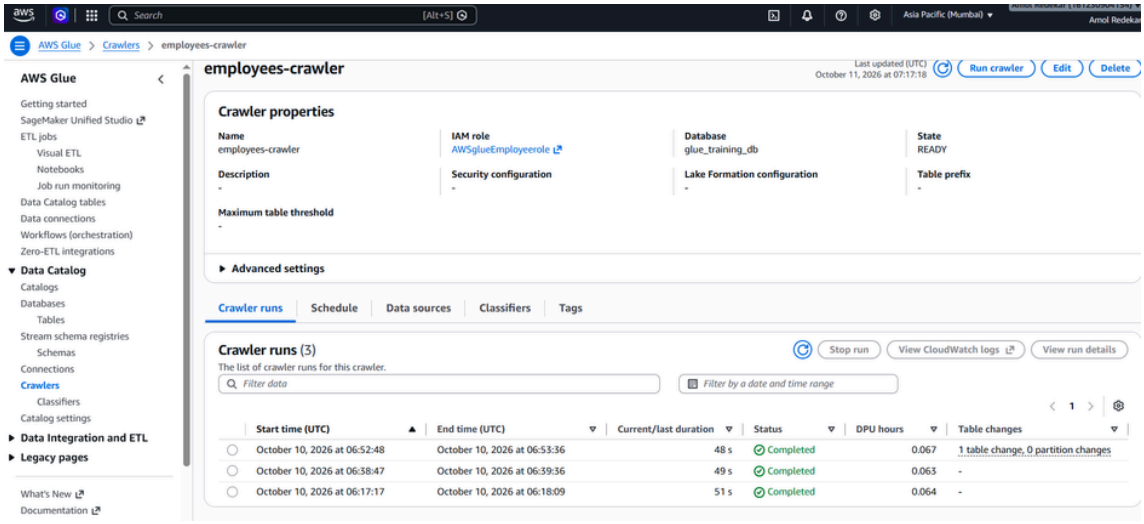
AWS Glue Database:

AWS Glue Data Catalog: database glue_training_db with its properties and tables



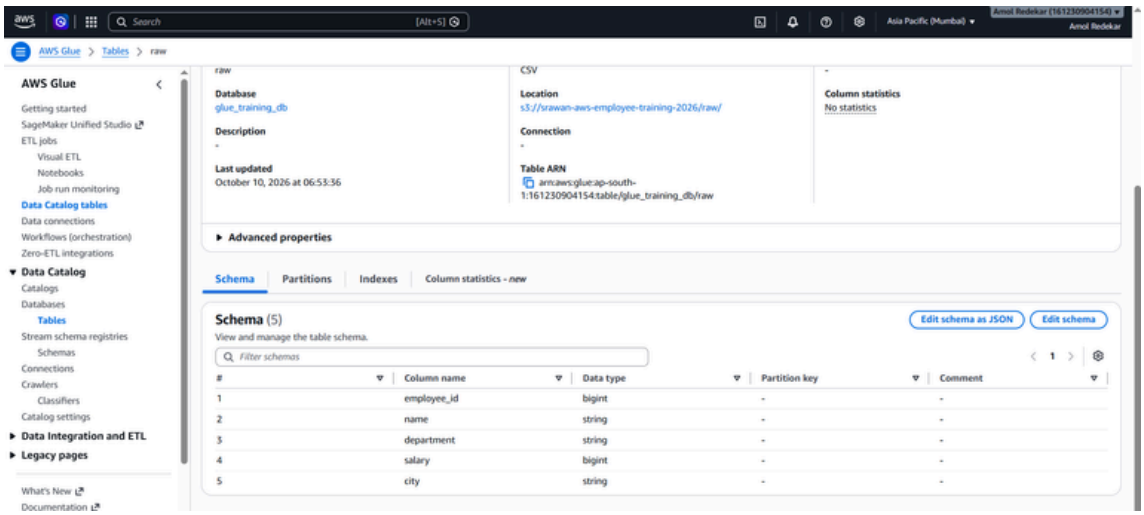
Glue Crawler:

Crawler employees-crawler: S3 data source (raw/ folder), IAM role AWSglueEmployeerole and target database glue_training_db



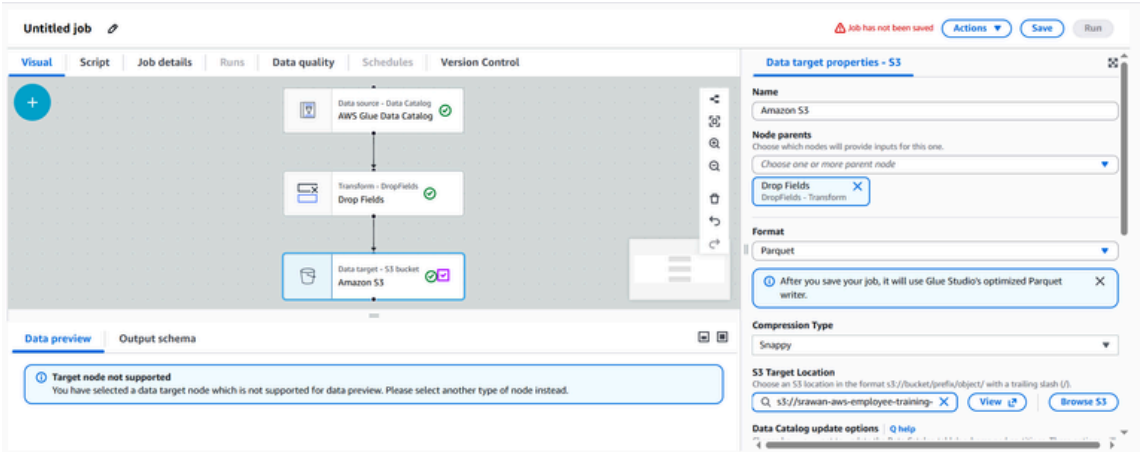
Glue Data Catalog:

Table raw created by the crawler in glue_training_db, with its schema and data types



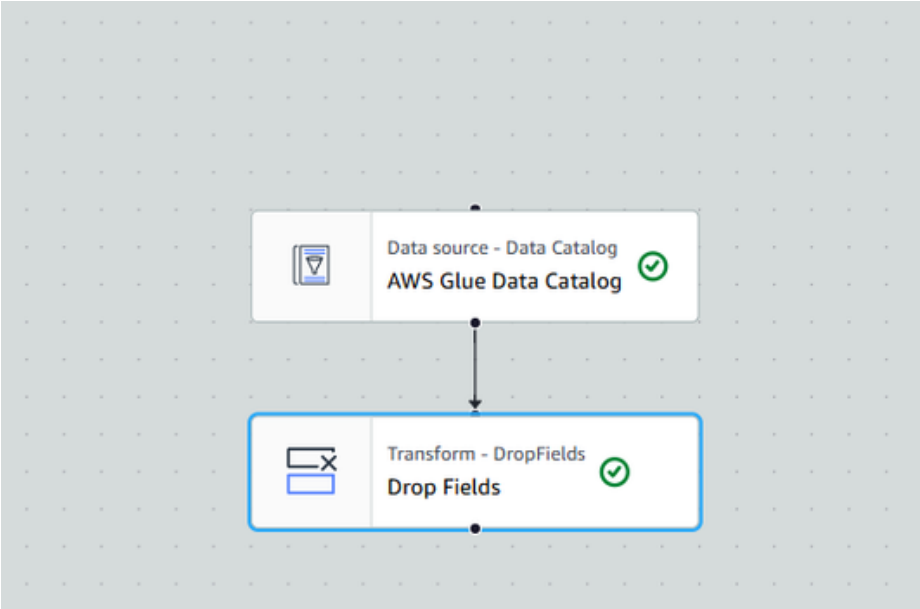
Glue Visual ETL:

Visual ETL job: Data Catalog source, Drop Fields transform and Amazon S3 target



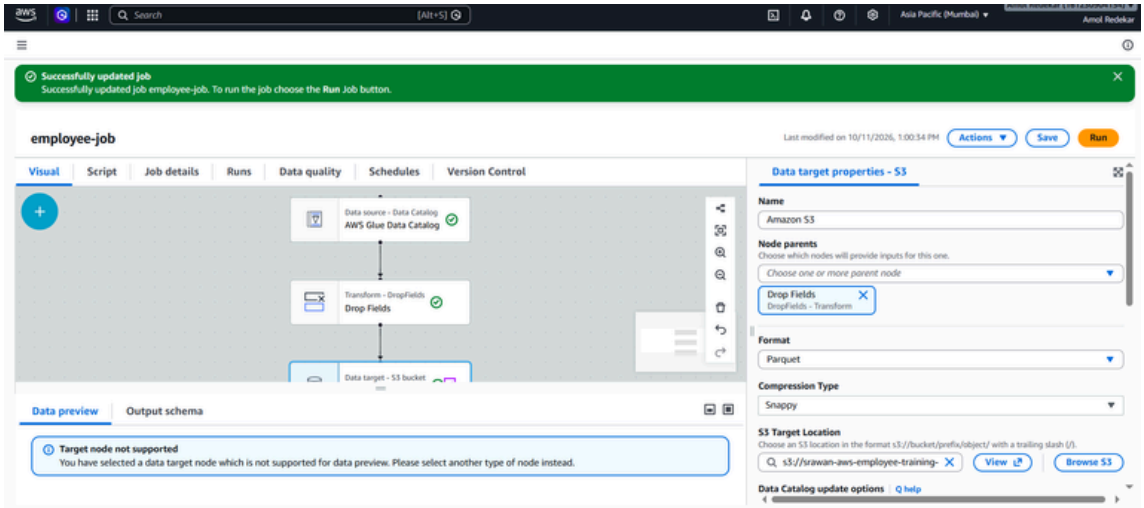
Glue Transformation:

Transformation: Drop Fields node connected to the AWS Glue Data Catalog source



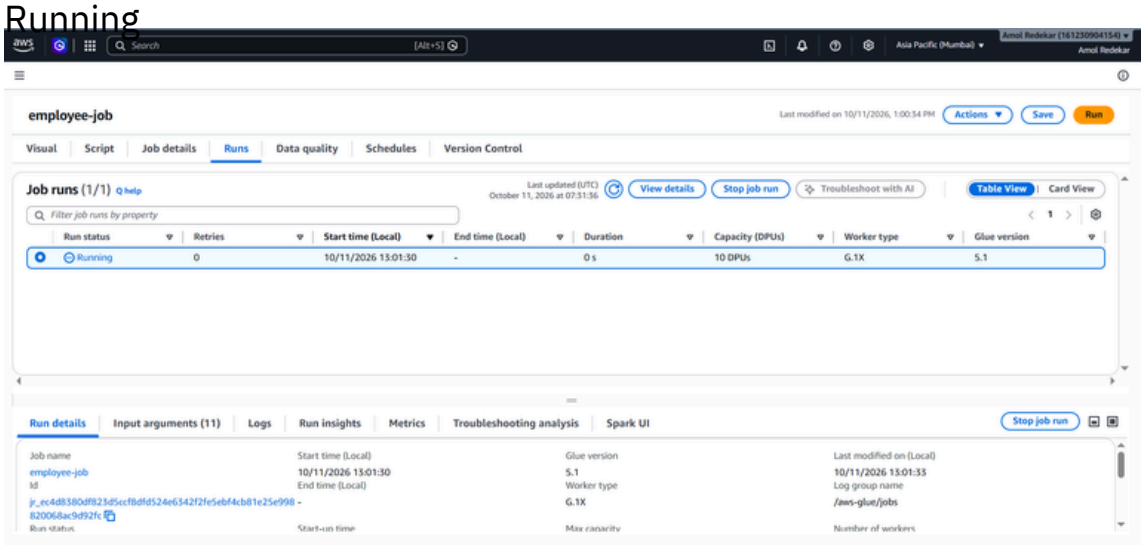
S3 Target Configuration:

S3 target configuration: Parquet format with Snappy compression and the S3 target location

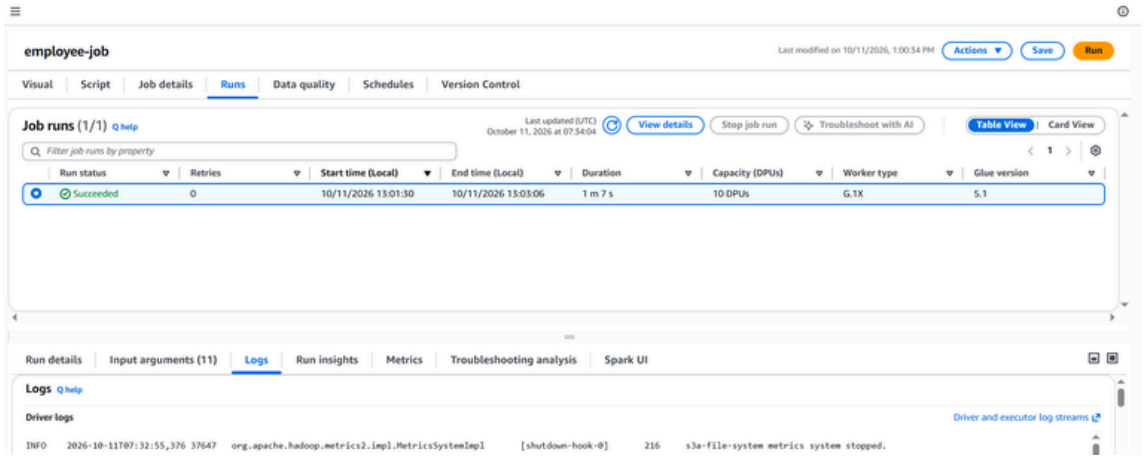


Glue ETL Job Execution:

Job employee-job started: run status Running



Job employee-job: run status Succeeded



Final S3 Output:

Output in the curated/ folder: a Parquet file generated by the job



Final Explanation: AWS Glue ETL Workflow

Workflow: S3 → Glue Crawler → *Glue Data Catalog* → Glue Visual ETL → *Transformation* → S3 Curated Data

1. Amazon S3: I uploaded the Employee CSV dataset (employee.csv) into the raw/ folder of my S3bucket.This is thesource data for the ETL workflow.

2. **AWS Glue Crawler:** The crawler scans the source data in S3, identifies its structure, detects column names and data types, and discovers the schema.

3. **AWS Glue Data Catalog*:*** The Data Catalog stores metadata about the dataset, including the table name, column names, data types, and source location. I created the bmw table in the glue_training_db database.

4. **AWS Glue Visual ETL*:*** I use the Visual ETL editor to design the data processing workflow by connecting source, transformation, and target nodes without writing all the code manually.

5. **Data Source*:*** The Visual ETL job reads the employee CSV data from the S3 raw/ folder or through the Data Catalog table, depending on the source configuration.

6. **Transformation*:*** The transformation node, such as Change Schema, allows me to select or rename columns and modify data types to prepare the dataset for further processing.

7. **S3 Curated Data*:*** The transformed data is written to the S3 curated/ folder in Parquet format. Parquet stores data in a columnar format that is useful for analytics.

8. **Final Result**: This workflow separates raw data from processed data, making the employee dataset more organized and ready for analytical tools such as Amazon Athena.

Conclusion*:* AWS Glue helps automate data discovery, manage metadata, transform datasets, and prepare structured data for analytics. The curated output is generated when the Visual ETL job runs successfully.

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